6061 Introduction to Deep Learning

Exercise 1

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1. **Installing Python.** When working with Python, it is often useful to install external packages. Since the required packages and their dependencies may vary based on your current project, it is highly recommended to install packages into a *virtual environment*. Using `uv` is an easy way to handle this.
   (a) Install `uv` by following the instructions at https://docs.astral.sh/uv/#installation.
   (b) Open a terminal, navigate to the directory where you want to keep the files for this course, and run the following commands:

```
# Create a subdirectory and initialize it as a Python project
uv init intro-dl
# Move into the project directory
cd intro-dl
# Install dependencies
uv add jupyter numpy
```

   This creates a virtual environment in `intro-dl/.venv/` and installs some useful Python packages. To install additional packages later, run `uv add PACKAGENAME` in the project directory. For more details, see the uv documentation.

2. (Optional) **Installing an IDE.** An integrated development environment (IDE) makes writing good code easier. For example, with automatic formatting your code always has the same style without you having to think about it. Other helpful features are static code analysis, which catches mistakes early, and sorting imports to keep them organized. We recommend Visual Studio Code (VS Code), which is free and has good support for Jupyter notebooks, in which we will write most of our code.
   (a) Download and install VS Code.
   (b) Open the Extensions view (`Ctrl+Shift+X`, or `Cmd+Shift+X` on macOS) and install the extensions *Python* and *Jupyter* (both by Microsoft).
   (c) Install the *Ruff* extension (by Astral Software), which formats your code.
   (d) Turn on formatting on save. Open the command palette (`Ctrl+Shift+P`, or `Cmd+Shift+P` on macOS), run *Preferences: Open User Settings (JSON)*, and add the following lines inside the outer curly braces:

```
"notebook.formatOnSave.enabled": true,
"[python]": {
    "editor.formatOnSave": true,
    "editor.defaultFormatter": "charliermarsh.ruff"
}
```

   If the file already contains other settings, separate the entries with commas.

   There are many more ways to customize VS Code, which you are welcome to try out.

3. **Using Python.** Download the Jupyter notebook for this exercise from Stud.IP and save it in your project directory `intro-dl`.

   With VS Code: Open the project directory via *File* → *Open Folder*, then open the notebook. Click *Select Kernel* in the top right corner, choose *Python Environments*, and select the environment in `.venv`. Without this step, the packages you installed with `uv` are not available in the notebook.

   Without VS Code: Open a terminal, navigate to the project directory, and run:

```
uv run jupyter lab
```

   This opens Jupyter in your web browser, where you can open the notebook.

   Work through the tasks in the notebook `01-hw-intro-py-tasks.ipynb`.